A1: Project Proposal Report

Shrimad Rajchandra Institute of Management and Computer Application
Uka Tarsadia university
MCA 2nd Semester
Course Name : Project Course Code: CS8034

Team Number	**MCA3_26**				
Project Title	**Ai TradeX – AI-Powered Investment & Stock Market Platform**				
Student guide interaction during	Day				Total Hours per week
	Time Duration				
Project Introduction	The stock market has become one of the most preferred investment options. However, many investors face difficulties in understanding market trends, selecting profitable stocks, managing investment risks, and analyzing large amounts of financial data. Most traditional trading applications provide only market information and trading facilities but lack intelligent decision-making support. **AI TradeX** is an AI-Powered Investment & Stock Market Platform designed to assist investors by using Artificial Intelligence, Machine Learning, and Data Analytics. The platform provides real-time stock prices, AI-generated Buy/Hold/Sell recommendations, portfolio management, market news, watchlists, and investment insights through a simple and user-friendly interface. The objective is to help both beginners and experienced investors make smarter investment decisions with the support of AI.				
Problem statements	Current stock market applications have several limitations: • Lack of AI-based investment recommendations. • Difficult for beginners to understand stock analysis. • Time-consuming manual market research.				

	• Limited portfolio analytics. • Lack of personalized investment suggestions. • Risk analysis is not easily understandable. • Market news is scattered across multiple platforms. • No integrated AI assistant for investment guidance.
System Need/ Objective	The primary objectives of AI TradeX are: • Develop an AI-powered investment platform. • Provide secure login using Google, Email, and Mobile OTP. • Verify user identity through KYC. • Display real-time stock market data. • Generate AI-based Buy, Hold, and Sell recommendations. • Help users manage investment portfolios. • Provide stock market analytics and visual dashboards. • Reduce investment risk through AI analysis. • Improve user decision-making. • Offer educational resources for beginners.
System Functionalities including reports	User Module
	• User Registration • Register & Login (Google, Email, Mobile OTP) • Profile Management • KYC Verification • Portfolio Management • Watchlist
	Investment Module
	• Real-Time Stock Prices • Stock Search • AI Stock Recommendations • Buy/Sell Simulation • Portfolio Tracking • Investment Calculator
	AI Module
	• AI Chat Assistant • Buy/Hold/Sell Prediction • Risk Analysis • Sentiment Analysis • Portfolio Optimization • Personalized Investment Suggestions

	Admin Module
	• User Management • KYC Approval • Dashboard • Market News Management • Reports • AI Monitoring
	Reports
	• User Registration Report • Portfolio Report • Investment Report • AI Recommendation Report • Profit/Loss Report • Risk Analysis Report • Top Gainers & Losers Report • Daily Active Users Report • Revenue Report
	Visual Analytics
	• User Growth Dashboard • Portfolio Distribution Chart • Stock Trend Line Chart • AI Recommendation Pie Chart • Sector Performance Dashboard • Risk Analysis Dashboard • Market Performance Dashboard
System Scope	The proposed system can be used by:
	• Beginner Investors • Experienced Traders • Students learning investments • Financial Advisors • Individual Investors
	The system provides:
	• AI-powered investment guidance. • Real-time stock market information. • Portfolio management. • Risk analysis. • Market analytics. • Cloud-based secure access. • Mobile application and web application support.

System Limitation	• AI cannot guarantee 100% accurate predictions. • Requires internet connectivity. • Depends on third-party stock market APIs. • Market volatility affects prediction accuracy. • KYC verification depends on external verification services. • Supports only stocks available through integrated APIs.
Tools and Technology	Frontend
	• Flutter (Android/iOS) • React.js / Next.js (Web) • Tailwind CSS
	Backend
	• Node.js • Express.js
	Database
	• MongoDB Atlas
	Artificial Intelligence
	• Groq API
	Authentication
	• JWT • Mobile OTP (Message Central)
	APIs
	• Stock Market API • News API • AI API
	Deployment
	• Render • Vercel • MongoDB Atlas • GitHub
	Development Methodology
	• Agile Software Development Model

Declaration:

We hereby declare that we have read the guidelines uploaded on the website for the above-said course and we shall follow those guidelines. We will not change the above-given details for the entire semester. I/We have chosen our team members by myself/ourselves. So, I/we will be responsible for team dispute, if any, during entire semester, which I/we will resolve by myself/ourselves.

We are well aware that project work must be prepared by the team members and must not be copied and (or) paid one. We acknowledge that the Institute has the authority for taking punitive actions against me for violation and/or non-compliance of the same.

Enrolment Number	**Name**	**Signature**
202504104610004	Jiya Patel	
202504104610093	Vidit Parekh	
202504104610095	Niral Patel	
202504104610136	Himanshu Thakkar	

Team members were made aware by me regarding plagiarism and its consequences.

Guide Name	**Signature**

Date:

Place : SRIMCA,UTU